

Project Proposal Template

Project Title

Real Estate Data Analytics in Latin America: Web Scraping and Forecasting

General Description

The real estate sector in Latin America is undergoing significant changes, with increasing demand for reliable and up-to-date information on property markets. This project focuses on collecting real estate data (rental and sale listings) from online platforms in Uruguay and Argentina through web scraping techniques. Students will then enrich the dataset with relevant macroeconomic indicators to better understand market dynamics. Finally, they will apply forecasting methods to predict future trends, combining technical skills with analytical thinking to produce actionable insights.

Project Scope

The purpose of the project is to introduce students to real-world data analytics challenges in the real estate domain. Students will first design and implement a scraping model to gather property listings, then clean and organize the data for analysis. By integrating macroeconomic variables such as inflation, GDP, or interest rates, students will learn how to contextualize real estate data within broader economic trends. The main objectives are to develop practical skills in programming and data science, while also strengthening critical thinking, teamwork, and communication through a final presentation of the results.

Student Requirements

Teamwork and collaboration in small groups

Full attendance and active participation in weekly meetings

High motivation and problem-solving mindset

Independent learning and research capabilities

Personal responsibility for assigned tasks and deadlines

Development Tools

Python (BeautifulSoup, Scrapy, Pandas, Scikit-learn)

Data sources: Real estate listing websites (Uruguay and Argentina), macroeconomic datasets (World Bank, IMF, national agencies)

GitHub for version control and code sharing

PowerPoint or Google Slides for presentation

Deliverables

List all required deliverables:

Specification document (project plan and methodology)

GitHub link to the code (scraping scripts, dataset, and forecasting model)

Presentation (slides summarizing process, findings, and insights)

Additional Notes

- Each project team may include up to 5 students.
- Please include your contact details at the end of the document for student inquiries:
 - Full name
 - Email address

For any questions, you may also contact: Shirany.hit@gmail.com